PROJECT 01B

Association Between Household Wealth and Adolescent Childbearing in Malawi: DHS 2017

1. Results

1.1 Analytical sample

The analysis included 828 adolescent girls aged 15–19 years. A complete-case approach was applied, and all three survey-weighted logistic regression models were fitted using the same analytical sample (N = 828).

1.2 Household wealth and adolescent childbearing

Three sequential survey-weighted logistic regression models were fitted to examine the association between household wealth and adolescent childbearing. Model 1 included household wealth only; Model 2 adjusted for age; and Model 3 additionally adjusted for educational attainment.

In Model 1, adolescents in the richest household wealth quintile had lower odds of having ever given birth compared with those in the poorest quintile (AOR = 0.292). Following adjustment for age in Model 2, the estimated association was AOR = 0.183. In Model 3, after further adjustment for educational attainment, the reported adjusted odds ratio for the richest quintile was 0.900 (95% CI: 0.191–0.900).

The Model 3 estimate and confidence interval are reproduced as supplied from the reported raw output. Because the point estimate is essentially identical to the upper confidence limit, this result warrants careful interpretation and verification against the saved model object before external publication.

1.3 Age and educational attainment

In the fully adjusted model (Model 3), age was positively associated with adolescent childbearing. Each additional year of age was associated with higher odds of having ever given birth (AOR = 3.046; 95% CI: 2.434–3.814).

Secondary education was associated with lower odds of having ever given birth compared with no education (AOR = 0.163; 95% CI: 0.049–0.540).

2. Table 2. Survey-weighted logistic regression results

Outcome: Ever having given birth. Reference categories: poorest household wealth quintile and no education. Age was modelled continuously.

Predictor	Model 1	Model 2	Model 3
Richest vs poorest household wealth	0.292	0.183	0.900 (0.191–0.900)
Age (per additional year)	—	—	3.046 (2.434–3.814)
Secondary vs no education	—	—	0.163 (0.049–0.540)
Analytical sample (N)	828	828	828

Values are adjusted odds ratios (AORs); 95% confidence intervals are shown in parentheses where supplied. Dashes indicate that the predictor was not included in that model. Confidence intervals for the Model 1 and Model 2 richest-versus-poorest estimates were not provided in the information used to prepare this report.

3. Summary of findings

The fully adjusted model showed associations between household wealth, age, educational attainment and adolescent childbearing. The reported association for the richest wealth quintile remained below the null value after adjustment, while older age was associated with higher odds and secondary education with lower odds of having ever given birth.

4. Interpretation and limitations

These findings describe associations and do not establish causal relationships. The cross-sectional nature of the DHS data limits conclusions about temporality. The complete-case approach may also introduce selection bias if adolescents with missing data differ systematically from those included in the analysis. Sparse outcome data in the richest wealth group may contribute to imprecision and should be considered when interpreting its estimate.

Reporting note: Before external submission, verify the Model 3 richest-group coefficient and confidence interval together from the same model output. The reported AOR (0.900) is virtually identical to the upper confidence limit (0.900), an unusual result that should not be explained as a confirmed mathematical boundary without further model diagnostics.